

Lecture 2: Single Person Decisions

JANUARY 21

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2.1 Decision Problems

2.1.1 Potato, Pasta, Rice ice breaker (10 minutes)

Ask the question, “Potato, Pasta, or Rice” with no context. Group them based on their response and let them introduce themselves. In each group, also have them ask about why they chose what they did, how they interpreted the question, and order themselves in a group based on how they interpreted the question.

2.1.2 Actions, Outcomes, and Preferences

We are going to set up the material for the first module today. Games are all around us, ask for examples for games. Today we start with single person games, or 1 agent games.

Definition 2.1 (Decision Problem) *A decision problem consists of three features:*

1. *Actions are all the alternatives from which the player can choose.*
2. *Outcomes are all the possible consequences that can result from any of the actions.*
3. *Preferences describe how the player ranks the set of possible outcomes, from most desired to least desired. The preference relation $\succsim$ describes the player's preferences, and the notation $x \succsim y$ means “ x is at least as good as y ”.*

Preference Relationships

Expanding on *preferences*, since the notation might be a little confusing. We can also have $y \succsim x$ means “ y is at least as good as x ”. This is a way to express the player's ranking of the possible outcomes. We also have the *strict preference relation* $x \succ y$ means “ x is strictly better than y ”. We lastly have $x \sim y$ which is the *indifference relationship* meaning “ x and y are equally as good”.

Definition 2.2 (The Completeness Axiom) *The preference relation $\succsim$ is complete: any two outcomes x, y out of the possible outcomes can be ranked by the preference relation, so either $x \succsim y$ or $y \succsim x$.*

This basically says that given two options, someone has to be able to rank them (including being indifferent). The player cannot be indecisive.

Definition 2.3 (The Transitivity Axiom) The preference relation $\succsim$ is transitive: for any three outcomes x, y, z from the outcome set, if $x \succsim y$, and $y \succsim z$, then $x \succsim z$.

Example 2.4 If I had to choose between potatoes, pasta, or rice, first I would say I prefer rice to pasta, and pasta to potatoes. This means without explicitly saying it, I prefer rice to potatoes by the transitivity axiom. This prevents an indecisive cycle.

Definition 2.5 (Rational Preference Relation) A preference relation that is both complete and transitive is called a rational preference relation.

Payoff Functions

If we have players with rational preferences, we can deal with payoff functions, which are nicer than preference relations.

Example 2.6 Imagine that you open a lemonade stand on your neighborhood corner. You have three possible actions: choose low-quality lemons (ℓ), which imply a cost of \$10 and a revenue from sales of \$15; choose medium-quality lemons (m), which imply a cost of \$15 and a revenue from sales of \$25; or choose high-quality lemons (h), which imply a cost of \$28 and a revenue from sales of \$35.

Thus the action set is $A = \{\ell, m, h\}$, and the outcome set is given by net profits and is $X = \{5, 10, 7\}$, where the action ℓ yields a profit of \$5, the action m yields a profit of \$10, and the action h yields a profit of \$7. Assuming that obtaining higher profits is strictly better, we have $10 \succ 7 \succ 5$. Hence you should choose alternative m and make a profit of \$10.

What if the problem doesn't depend on money or profit? This is why we define payoff functions

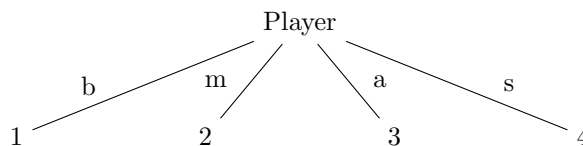
Definition 2.7 (Payoff function) A payoff function u represents the preference relationship $\succsim$ if for any two outcomes x, y from the outcome set X , then $u(x) \geq u(y)$ if and only if $x \succsim y$.

Remark 2.8 The payoff function u assigns higher values to higher-ranked outcomes, but the payoff values themselves don't really matter, as long as they keep the ordering.

Proposition 2.9 If the set of possible outcomes X is finite, then any rational preference relationship over X can be represented by a payoff function.

Decision Trees

Imagine that, in addition to Apple Jacks (a) and Bran Flakes (b), your breakfast options include a muffin (m) and a scone (s). Your preferences are given as $s \succ a \succ m \succ b$. (Recall that we now consider preferences over outcomes as directly over actions.) Consider the following payoff representation: $v(s) = 4, v(a) = 3, v(m) = 2$, and $v(b) = 1$.



2.2 The Rational Choice Paradigm

Definition 2.10 (Rational Choice Assumptions) *The player fully understands the decision problem by knowing:*

1. *all possible actions, A .*
2. *all possible outcomes, X .*
3. *exactly how each action affects which outcomes will materialize, and*
4. *the payoff function over the outcomes.*

Remark 2.11 *We say outcome $x(a)$ is the outcome resulting from action a , then the payoff from action a is given by $v(a) = u(x(a))$, the payoff function from action $x(a)$.*

Definition 2.12 (Rational) *We say that an agent faced with a decision problem is rational if they choose a^* from the set of all actions such that $v(a^*) \geq v(a)$ for all of the options that could have been chosen.*

So far we have only talked about finite action set examples, e.g. “should I do x or y ?”. Now we will consider an example with a continuous action space. Imagine you are studying for an exam and need to really lock in. You are thinking about drinking energy drinks to help you really focus. If you drink some, you will be focused, energized, and able to study, but if you drink too much, you will feel sick, anxious, nauseous. You have a 24 oz can of Monster, so your action set is $A = [0, 1]$, where $a \in A$ is the percent of the can you choose to drink. Your preferences are represented by the following payoff function over actions: $v(a) = 2a - 4a^2$, which is depicted in figure 2.2.

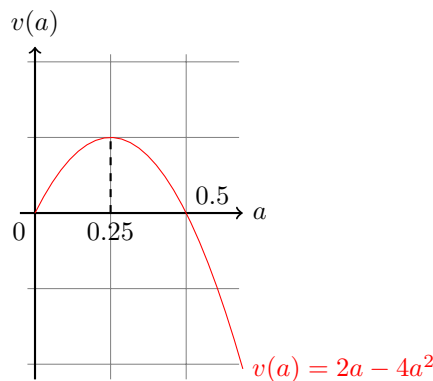
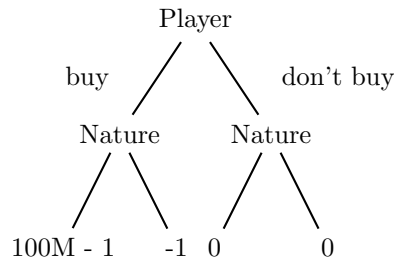


Figure 2.1: payoff value from drinking different percentages of the 24 oz can.

As you can see, some Monster is better than no Monster (0.1 or 10% of the can gives you some positive payoff, while drinking nothing gives you zero), but drinking a whole can will be worse than not drinking at all ($v(1) = -2$). How much Monster should you drink? Your maximization problem is

$$\max_{a \in [0,1]} 2a - 4a^2.$$



Taking the derivative of this function and equating it to zero to find the solution, we obtain that $2 - 8a = 0$, or $a = 0.25$ or 25%, which is 6 oz of Monster. We also must check our endpoints of $\{0, 1\}$, but we already noted that $v(0) = 0$, and $v(1) = -2$, so since $v(0.25) > 0$, we have found the value that maximized our payoff, and our choice as a rational consumer.

2.3 Uncertainty and Time

How does a rational person consider uncertainty. If I buy a lottery ticket, I can either win or not win, but these probabilities aren't equal.

Let's say that a lottery ticket is \$1, and the payout is \$100 million dollars.

Probability refresher: By the definition of a probability distribution over elements in X , the probability of each outcome cannot be a negative number, and the sum of all probabilities over all outcomes must add up to 1.

Examples include flipping a coin and rolling a dice.

Some outcomes are certain, like the breakfast situation. If I chose the scone, I would know that I get a payoff of 4. There is no chance I would get anything else. There is nothing up in the air.

However, sometimes there is a probability distribution over outcomes conditional on chosen action.

Definition 2.13 (Expected Payoff from the lottery) Let $u(x)$ be the player's payoff function over outcomes in $X = \{x_1, x_2, \dots, x_n\}$, and let $p = (p_1, p_2, \dots, p_n)$ be a lottery over X such that p_k is the probability of x_k being picked. Then we define the player's expected payoff from the lottery p as

$$\mathbb{E}[u(x)|p] = \sum_{k=1}^n p_k u(x_k) = p_1 u(x_1) + p_2 u(x_2) + \dots + p_n u(x_n)$$